



Course Overview

23CSE351 Foundations of Data Science Lecture 0

Course Details

Course Code :- **23CSE351**

Course Title :- **Foundations of Data Science**

Course L-T-P :- **3-0-0-3**

Course Credits :- **3**

Course Type :- **Elective based Course**

Course Objective

- The objective is to teach primary tools for exploration, visualizations and descriptive statistics, for prediction are machine learning and optimization, and for inference are statistical tests and models.
- Through understanding a particular domain, the students learn to ask appropriate questions about their data and correctly interpret the answers provided by inferential and computational tools.

Course Outcomes

- **CO1:** Understand the statistical foundations of data science.
- **CO2:** Apply pre-processing techniques over raw data to enable further analysis
- **CO3:** Conduct exploratory data analysis and create insightful visualizations to identify patterns.
- **CO4:** Identify machine learning algorithms for regression/classification tasks and to get into insights.
- **CO5:** Analyze the degree of certainty of predictions using statistical tests and models.

Course Syllabus

- **Unit 1**

- Introduction to Data Science, Causality and Experiments, Data Preprocessing: Data cleaning, Data reduction, Data transformation, Data discretization. Exploratory Data Analysis in python: Visualizing categorical data, numerical data, summary statistics of data, overlaid graphs. Random Variables: Random variables, Functions of Random variables Probability Distributions: Discrete and continuous distributions, Sampling: Sampling Concepts, The Central Limit Theorem and Applications. Sample Means and Sample Sizes.

- **Unit 2**

- Descriptive statistics: Central tendency, dispersion, variance, covariance, kurtosis, five-point summary, Distributions, Bayes Theorem, Error Probabilities; Permutation Testing, Hypothesis and Inference: P-Values, Hypothesis Testing, Assessing Models, Decisions and Uncertainty, Comparing Samples, Chisquared Test, A/B Testing.

- **Unit 3**

- Linear Regression: Building the regression model - Least square line, Predictions using regression models – Uncertainties in regression coefficients, checking assumptions and transforming data, web scrapping, Introduction to Data Visualization Tools: Tableau, PowerBI.

Text Books/References

- **Text Book:**
- Ani Adhikari. John DeNero, Computational and Inferential Thinking: The Foundations of Data Science. GitBook, 2019.
- **Reference(s):**
- Shmueli G, Bruce PC, Yahav I, Patel NR, Lichtendahl Jr KC. Data mining for business analytics: concepts, techniques, and applications in R. John Wiley & Sons; 2018..
- Schutt R, O'Neil C. Doing data science: Straight talk from the frontline. First Edition, O'Reilly Media, Inc.; 2013.

Course Evaluation Policy

- Internal-70
 - MidTerm-20
 - 2 Quizzes(5 mark each)-10
 - Data camp Submission-10
 - Data camp Quiz-10
 - Case Study-20 marks
- External-30